# Arista Access Node QoS Congestion Event Troubleshooting and Resolution Procedure (BAS-ACC-10)

**Node:** BAS-ACC-10 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_9525

## Description

The described failure domain, specifically ALARM_9525, is a Minor Severity QoS congestion event on the Arista Access Node BAS-ACC-10 at the Basavanagudi POP. This event is primarily attributed to Microbursts, interface tail drops, and QoS queue starvation, which are indicative of a transient congestion anomaly within the node's egress interfaces. The root cause of this issue lies in the dynamic nature of network traffic patterns, coupled with the Arista Access Node's configuration for Quality of Service (QoS) management, which prioritizes traffic based on pre-defined policies. When traffic bursts or other anomalous patterns occur, the Access Node's QoS configuration may lead to congestion on specific interfaces, resulting in alarm conditions that trigger the ALARM_9525 alarm code. Furthermore, inadequate buffer allocation or misconfigured QoS policies can exacerbate this issue, leading to interface tail drops and queue starvation. In order to effectively troubleshoot and resolve this issue, it is essential to analyze the Access Node's configuration, inspect the traffic patterns, and verify the integrity of the QoS policies.

## Pre-requisite Checks

- Verify that the Basavanagudi POP has been physically accessed by authorized personnel.

- Ensure that the redundant link status is checked and functioning as expected.

- Confirm that the Power System (PS) is operational and within normal parameters.

- Verify that the Environmental Monitoring System (EMS) is functioning correctly and within normal parameters.

- Check the network connectivity to the Access Node BAS-ACC-10 via the management interface.

## Potential Root Causes

| Cause | Probability |
|---|---|
| Misconfigured QoS policies leading to inadequate buffer allocation. | High |
| Inadequate buffer allocation due to insufficient configuration. | Medium |
| Traffic bursts or anomalous patterns causing interface congestion. | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show interfaces status, Expected Output: Interface Status:
Up/Down, Speed: xx Gbps, Duplex: Full/Half, Port-Channel: On/Off,
```

```
Description: This command is used to verify the status of the
interfaces on the Access Node BAS-ACC-10. The expected output will
indicate the status of each interface, including any errors or
anomalies that may be contributing to the congestion event.

Command: show qos policer, Expected Output: Policer Status: Up/Down,
Rate: xx Mbps, Burst: xx packets, Action: Drop/Mark, Description: This
command is used to inspect the QoS policies configured on the Access
Node BAS-ACC-10. The expected output will provide information on the
policers, including their status, rate, burst size, and action.

Command: show buffer, Expected Output: Buffer Status: xx packets, xx
bytes, xx drops, xx overflows, Description: This command is used to
analyze the buffer allocation on the Access Node BAS-ACC-10. The
expected output will provide information on the buffer status,
including the number of packets, bytes, drops, and overflows.
```

## Resolution Steps

Step 1: Step: Verify and adjust QoS policies, Command: configure qos policer rate burst , Description: This step is used to adjust the QoS policies on the Access Node BAS-ACC-10. The command will update the policer configuration to adjust the rate and burst size, which will help alleviate congestion on the interfaces.

Step 2: Step: Verify and adjust buffer allocation, Command: configure buffer allocate , Description: This step is used to adjust the buffer allocation on the Access Node BAS-ACC-10. The command will update the buffer allocation to increase the buffer size, which will help alleviate congestion on the interfaces.

Step 3: Step: Verify and apply traffic shaping, Command: configure qos traffic-shape , Description: This step is used to apply traffic shaping on the Access Node BAS-ACC-10. The command will update the traffic shaping configuration to limit the traffic rate and burst size, which will help alleviate congestion on the interfaces.

## Post-Check Verification

```
Command: show interfaces status, Expected Output: Interface Status:
Up/Up, Speed: xx Gbps, Duplex: Full/Full, Port-Channel: On/On,
Description: This command is used to verify that the interfaces are
functioning correctly after the resolution steps have been applied.
The expected output will indicate that the interfaces are up and
functioning correctly.

Command: show qos policer, Expected Output: Policer Status: Up/Up,
Rate: xx Mbps, Burst: xx packets, Action: Drop/Mark, Description: This
command is used to verify that the QoS policies have been adjusted
correctly after the resolution steps have been applied. The expected
output will provide information on the policers, including their
status, rate, burst size, and action.

Command: show buffer, Expected Output: Buffer Status: xx packets, xx
bytes, 0 drops, 0 overflows, Description: This command is used to
verify that the buffer allocation has been adjusted correctly after
the resolution steps have been applied. The expected output will
provide information on the buffer status, including the number of
packets, bytes, drops, and overflows.
```

## Roll-back Procedures

- Step: Revert QoS policies, Command: configure qos policer rate burst , Description: This step is used to revert the QoS policies on the Access Node BAS-ACC-10 to their original configuration. The command will update the policer configuration to restore the original rate and burst size, which will help resolve any issues caused by the resolution steps.

- Step: Revert buffer allocation, Command: configure buffer allocate , Description: This step is used to revert the buffer allocation on the Access Node BAS-ACC-10 to its original configuration. The command will update the buffer allocation to restore the original buffer size, which will help resolve any issues caused by the resolution steps.

- Step: Revert traffic shaping, Command: configure qos traffic-shape , Description: This step is used to revert the traffic shaping configuration on the Access Node BAS-ACC-10 to its original configuration. The command will update the traffic shaping configuration to restore the original rate and burst size, which will help resolve any issues caused by the resolution steps.